

A Bayesian framework to model transmissible cancer dynamics within soft-shell clam (*Mya arenaria*) populations



Carissa Mayo^{1,3}, Robert Noble², Michael Metzger³
¹ Quantitative Ecology and Resource Management, University of Washington
² Department of Mathematics, City St. George's, University of London
³ Pacific Northwest Research Institute



Background

Bivalve Transmissible Neoplasia (BTN)

- Transmissible cancer occurs in a few species, including Tasmanian devils, dogs, and several bivalve species.
- In soft-shell clams, a leukemia-like cancer was confirmed to be transmissible in 2015 and is spread through waterborne cancer cells¹
- Infected clams release cells that can survive for weeks, especially in cold temperatures, leading to seasonality in disease prevalence²
- Seasonal disease patterns are observed in wild populations, but the mechanism generating these patterns remain unclear.



Soft-shell Clam

Marine Disease Modeling

- Most disease models are built for terrestrial systems, but marine diseases follow different dynamics
- Environmental factors like temperature influence pathogen survival and transmission, shaping seasonal disease dynamics

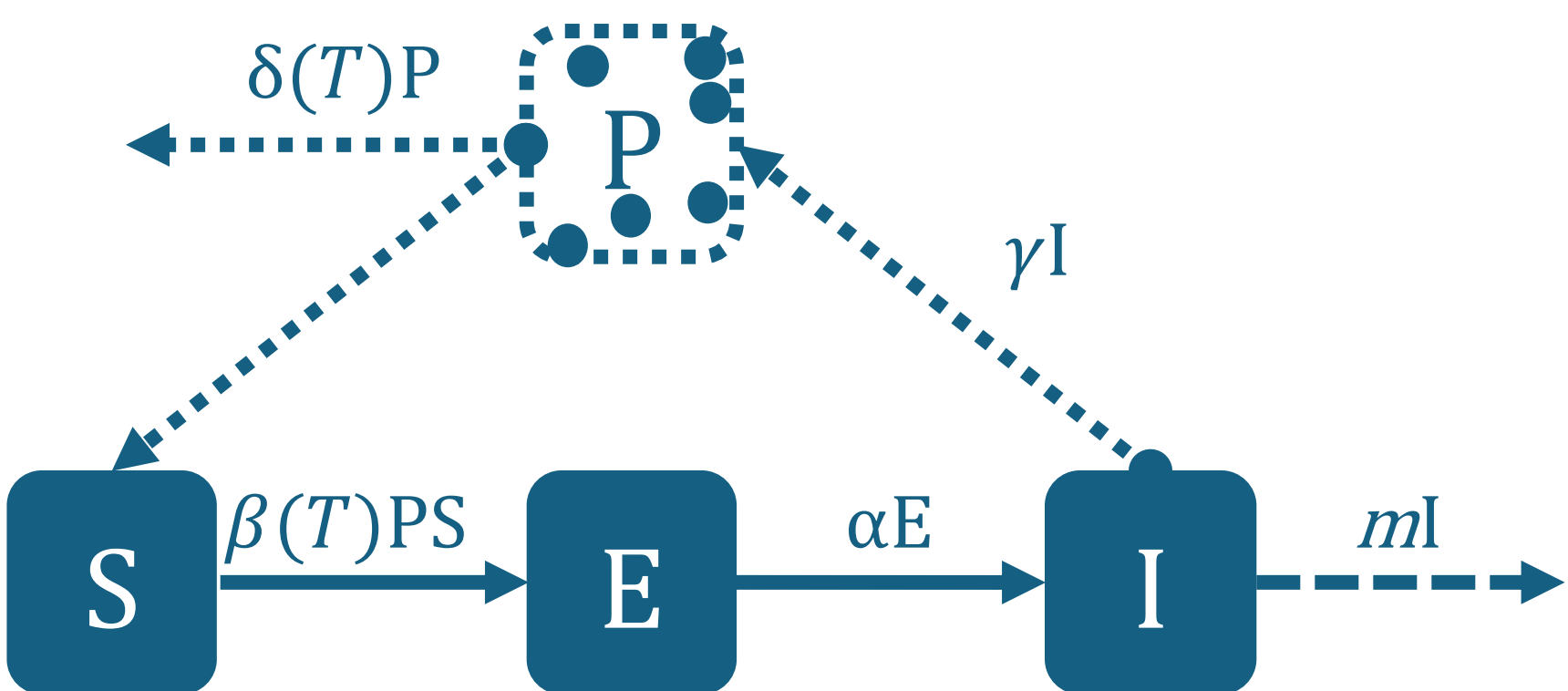
Bayesian Approaches in Disease Modeling

- Bayesian methods combine lab data and field observations.
- We use Stan³, a probabilistic programming language, to:
 - Efficiently fit ordinary differential equation-based models
 - Estimate key transmission and progression parameters
 - Quantify uncertainty in predictions

Key Objectives

- Develop a temperature-dependent SEIP model for environmental cancer transmission in soft-shell clams
- Estimate transmission, progression, and particle emission rates using Bayesian inference
- Test the hypothesis that incorporating seasonal temperature effects allows the model to reproduce observed trends in disease dynamics

Model Structure



Adapted from a marine infectious disease model⁴:

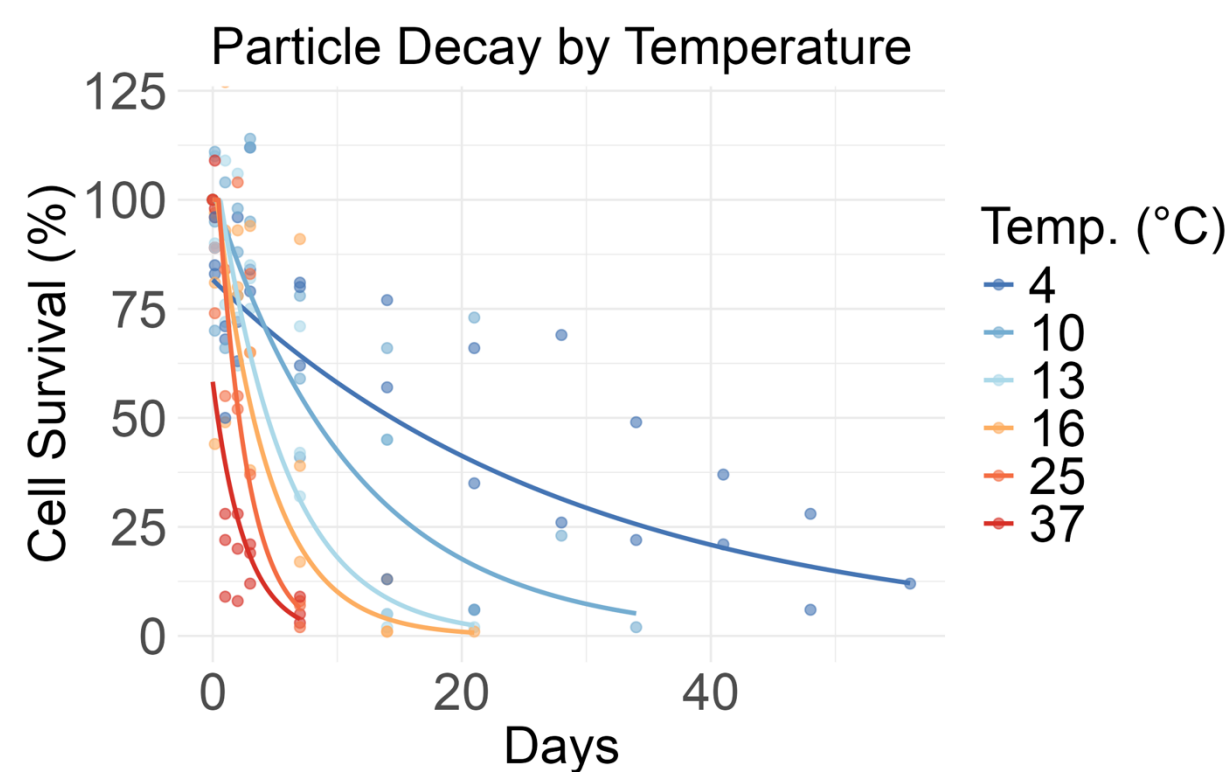
- Susceptible clams (S) become exposed (E) via contact with cancer cells in the water at rate β .
- Progression rate from exposed to infectious is α .
- Infectious clams (I) release cancer cells (particles, P) into the environment at rate γ and leave the system at mortality rate m .
- Particles leave the system with a decay rate δ .

We run and compare results from 4 different models to assess where temperature best fits into the system at this site:

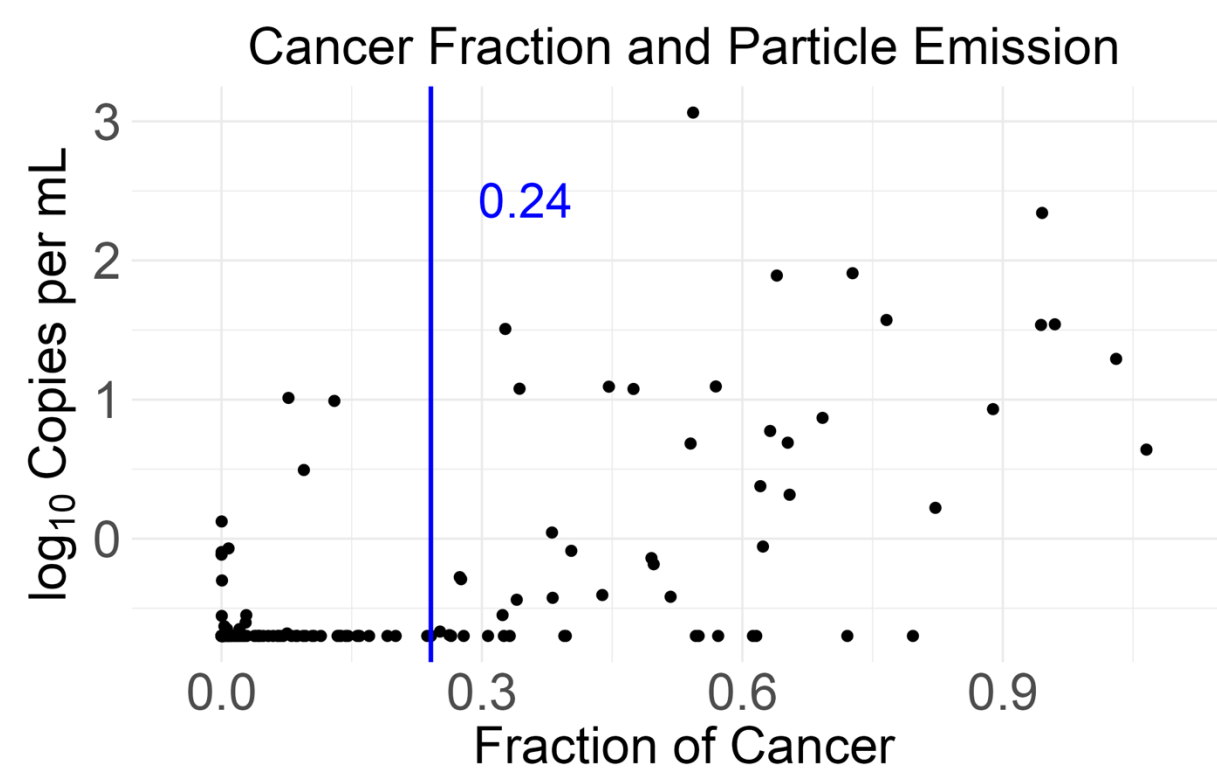
Model	Description
1	No temperature dependence
2	Temperature dependence in particle decay: $\delta(T)$
3	Temperature dependence in progression rate: $\beta(T)$
4	Both temperature dependent parameters: $\delta(T)$ and $\beta(T)$

Parameter Priors and Survey Data

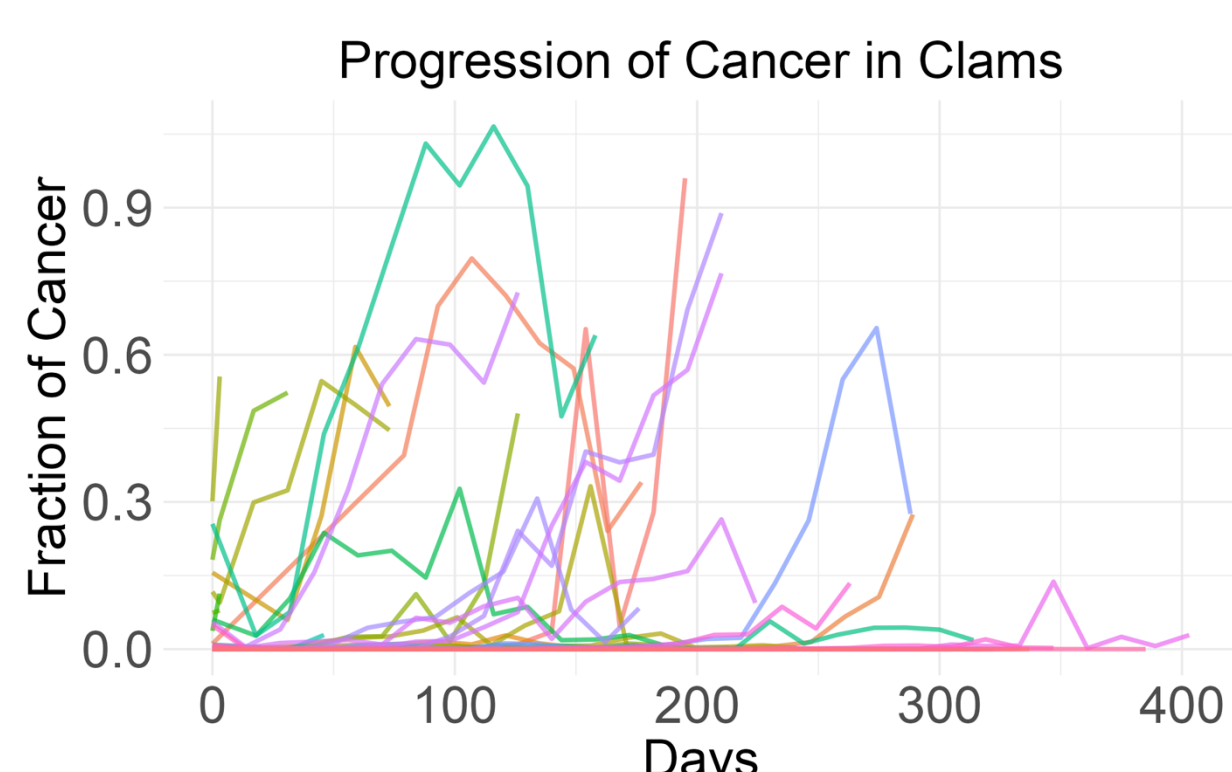
Laboratory Data Used to Inform Model Parameters



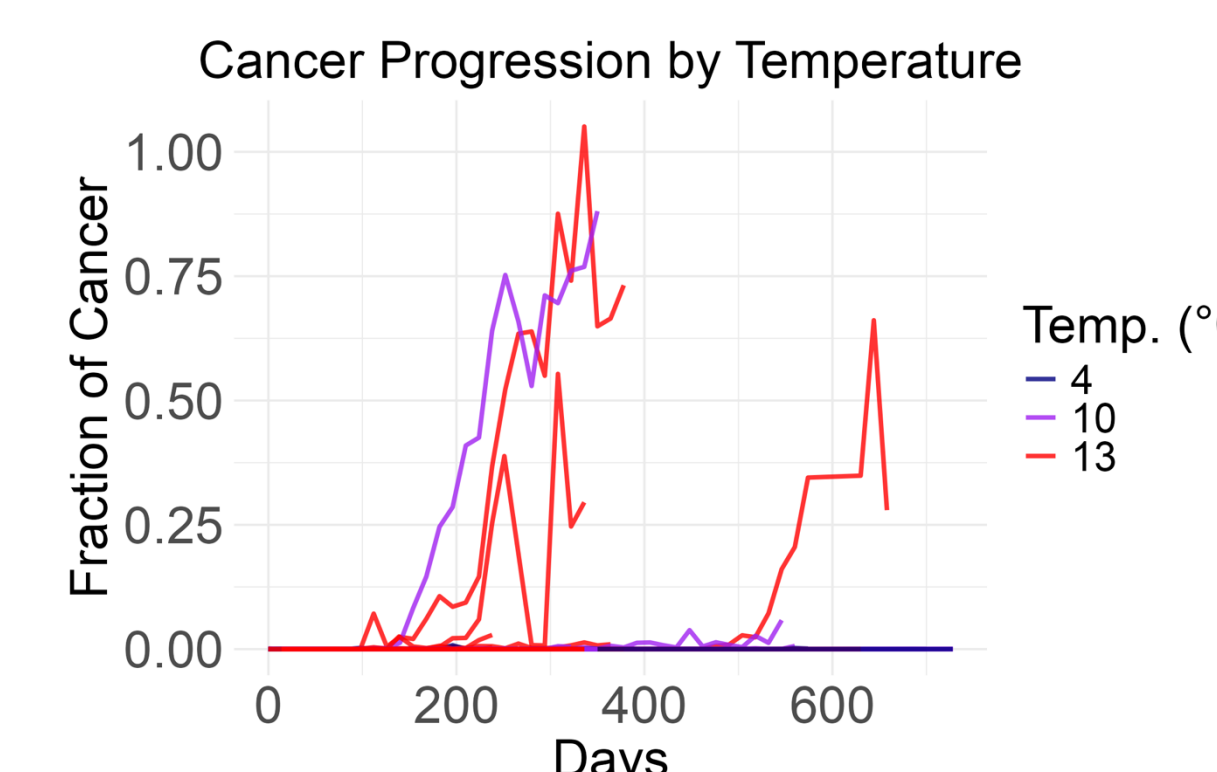
Plot A: Cancer cell decay at different temperatures informs δ and its temperature dependence.



Plot B: Cancer fraction vs. particle release per clam. A changepoint signals infectiousness; later data informs γ .



Plot C: Longitudinal data from clams. Time to 0.24 informs α ; time from 0.24 to death informs m .

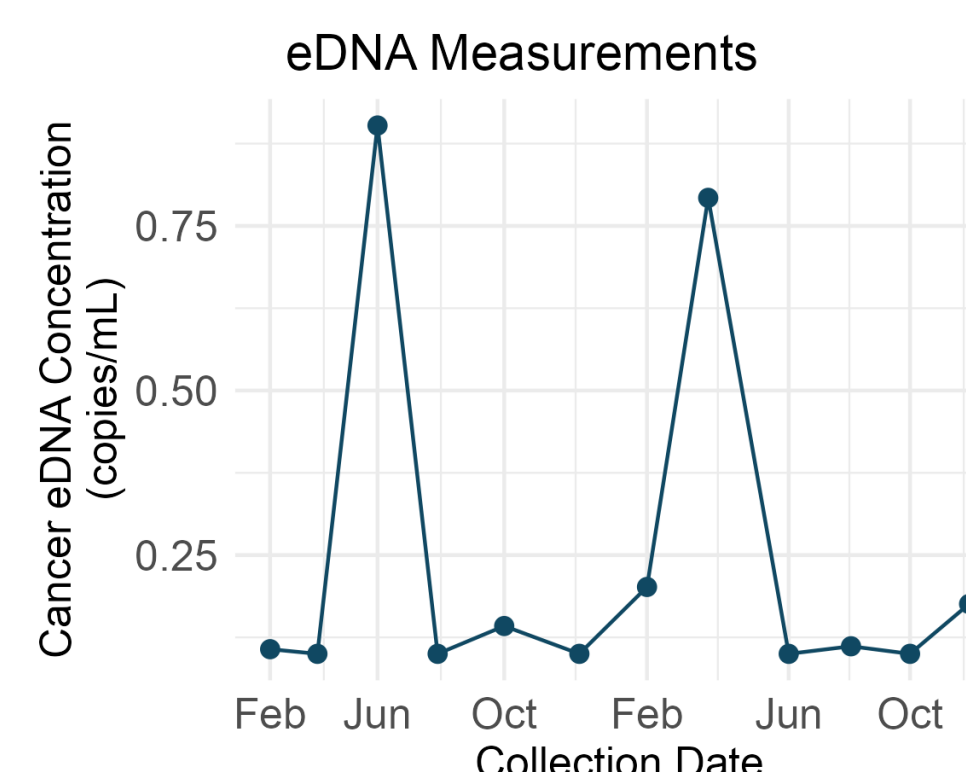
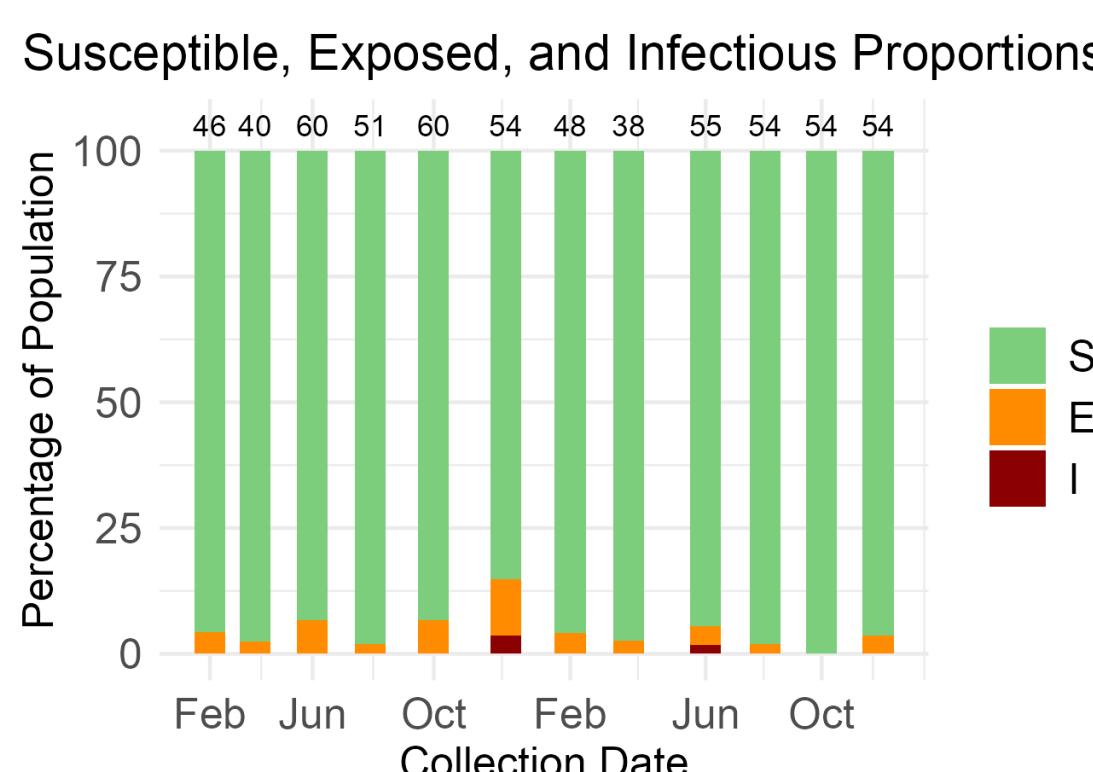


Plot D: Longitudinal data from clams kept at various temperatures informs $\beta(T)$ temperature dependence.

Observational Data

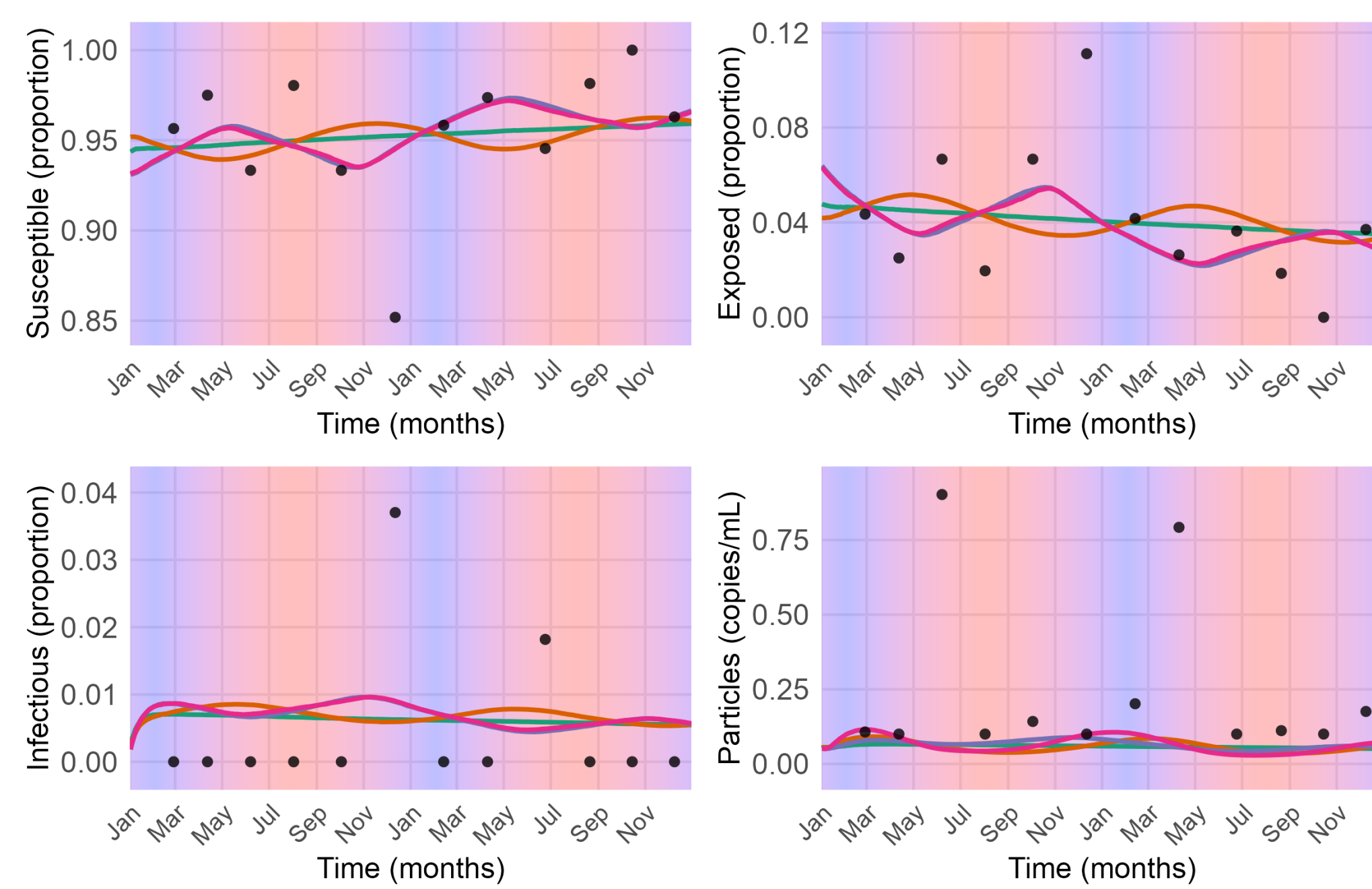
- Field surveys conducted bi-monthly along the East Coast.
- ~50 clams per time point.
- eDNA from seawater measures cancer cell concentration.

Survey Results from Forest River, MA (Feb. 2023 - Dec. 2024)



Results

SEIP Median Trajectories by Compartment and Model

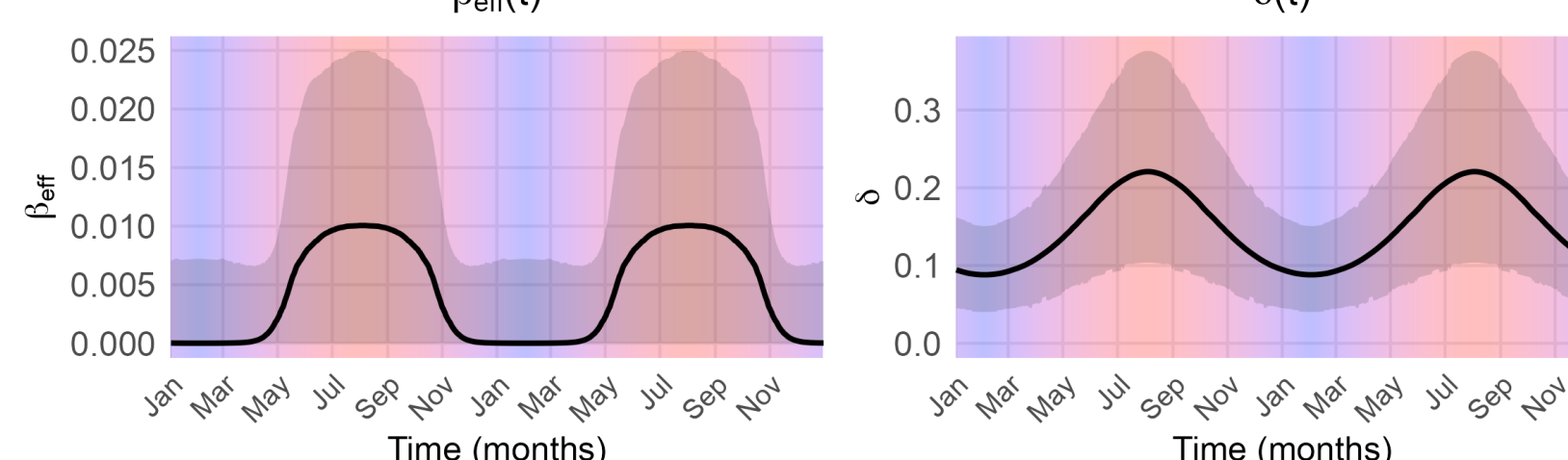


Disease progression ($\beta(T)$) is the dominant seasonal driver in this site.

Captures overall disease dynamics and the main seasonal trend, but misses some observed patterns.

Temperature dependent parameter estimates align with lab data relationships.

Posterior Temperature Dependence of β_{eff} and δ Over Time

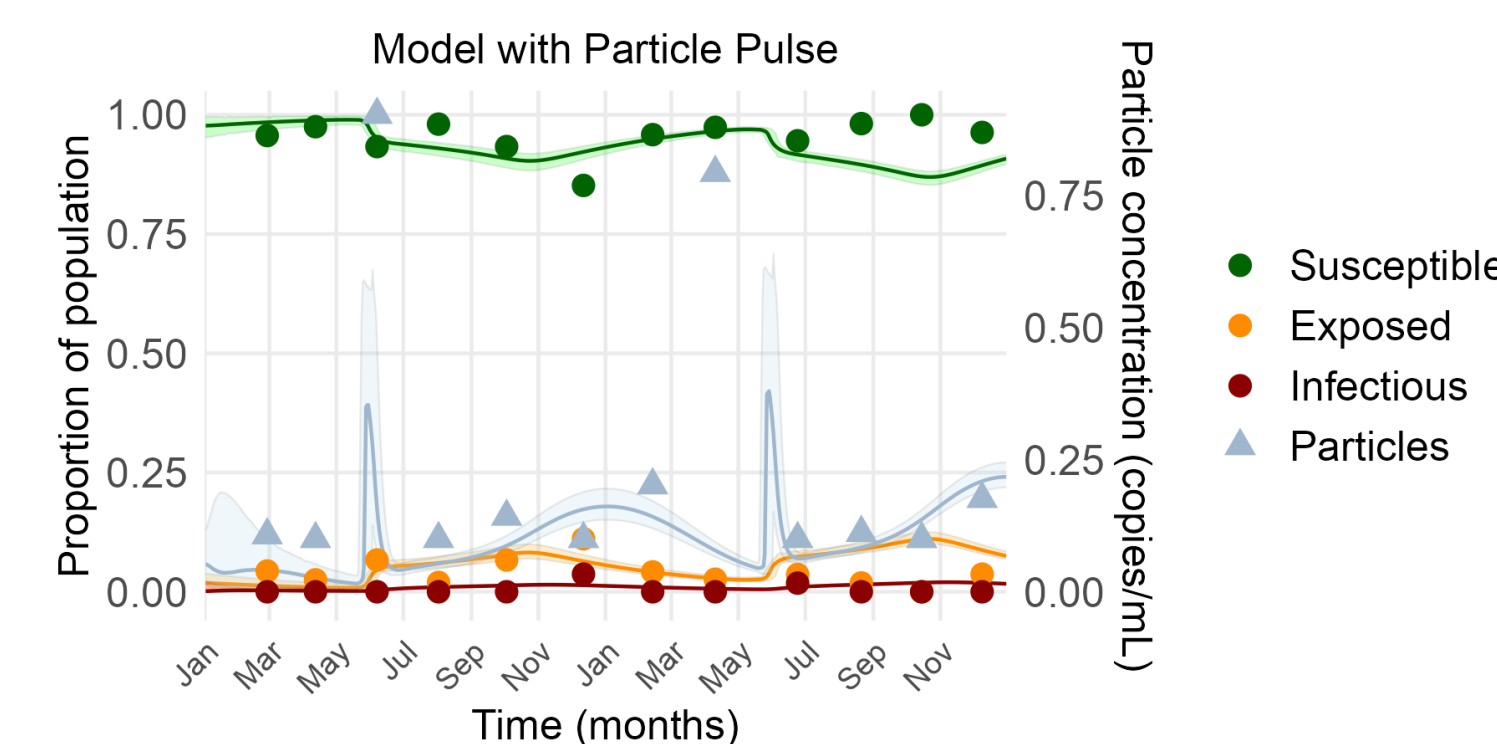


Discussion

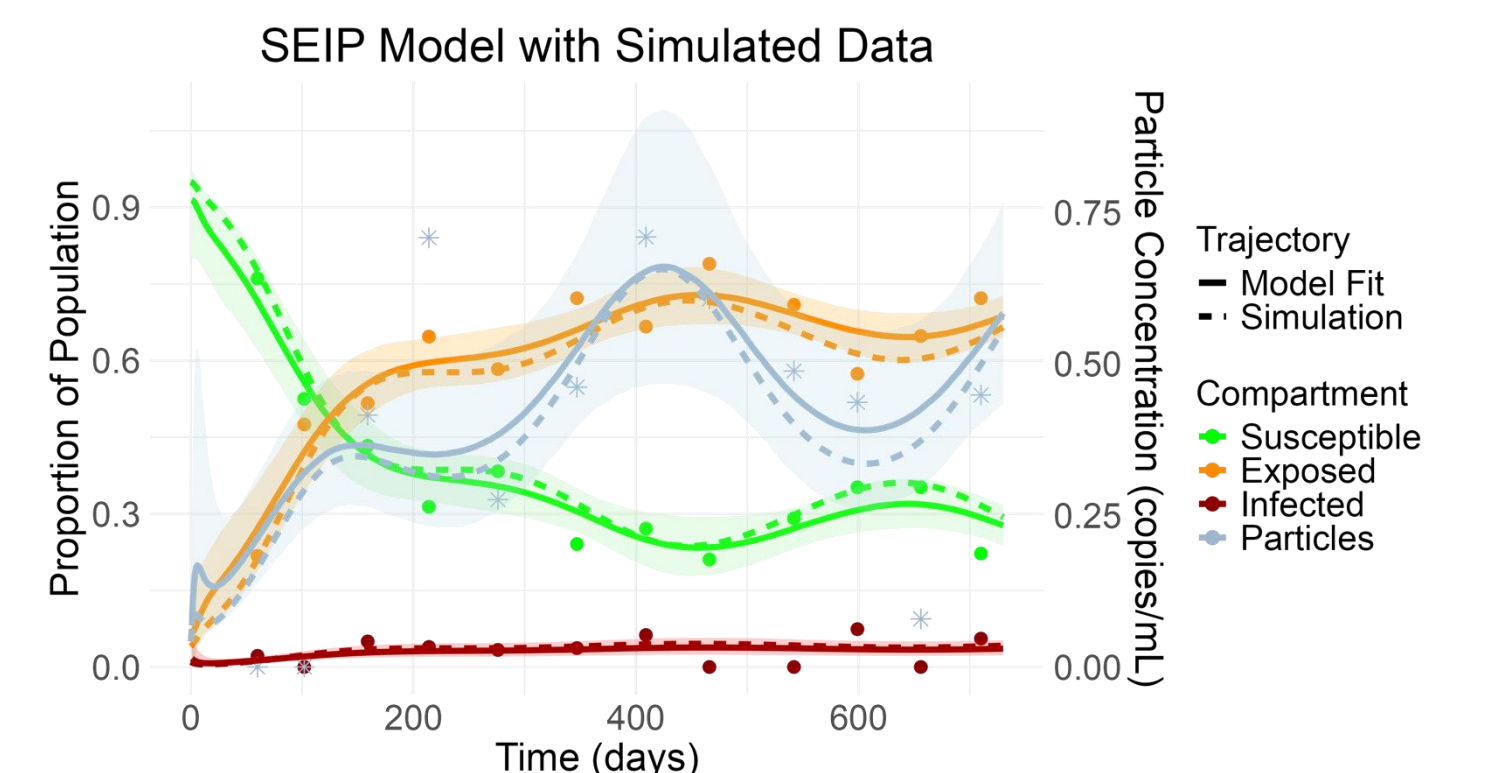
Discrepancies between the model and field data may be due to:

- Observation error/detection limits
- Biological processes not included in the model (e.g., spawning)

Theoretical experiment showing a short, seasonal particle pulse can capture seasonal patterns not in models 1-4 results.



Simulation tests were conducted to determine whether the model and observation process can recover known parameter values under realistic sampling and noise. Results showed reliable recovery of model parameters, supporting the robustness of inference at this site.



Future Work

- Generate predictions for 2025 and validate against future survey data.
- Increase sampling frequency to improve resolution.
- Apply to additional sites to identify site differences.
- Extending model mechanisms for site-specific attributes where temperature forcing alone is insufficient.



Acknowledgments

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